

BONIFACE B. SANKWA,  
22100934340091.

MBEYA UNIVERSITY OF SCIENCE AND TECHNOLOGY

TEST: ET 8212 – ELECTROMAGNETIC THEORY I

Time: 60 minutes

31<sup>st</sup> May 2024

- Answer ALL questions
- Write all answers in the space provided in the question paper.

Question One

- With illustrative equation, state Divergence theorem (2Marks)
- Calculate the circulation of  $A = \rho \cos \phi a_\rho + z \sin \phi a_z$  around the edge L of the wedge defined by  $0 \leq \rho \leq 2, 0 \leq \phi \leq 60^\circ, z = 0$  (4Marks)

Question Two

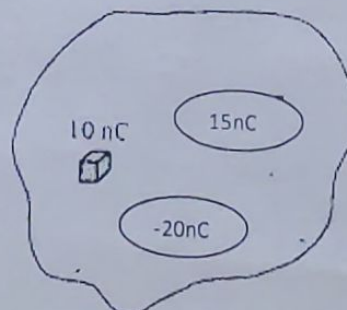
- What is Electrical potential? (2Marks)
- A point charge of  $-4 \mu\text{C}$  is located at  $(2, -1, 3)$ , Find the potential at  $(1, 0, 1)$  assuming zero potential at infinity (4Marks)

Question Three

- With illustrative equation, state coulomb's Law (2Marks)
- Draw the electric field line pattern for Electric Dipoles (2Marks)
- Point charge of  $1 \text{ mC}$  is located at  $(3, 2, -1)$ . Calculate the electric force on a  $10\text{-nC}$  charge located at  $(0, 3, 1)$  and the electric field intensity at that point. (6Marks)

Question Four

- With illustrative equation, state gauss's law (2Marks)
- Given the figure below



- Calculate the total charge enclosed (3Marks)
- Calculate the total outward flux leaving the surface (3Marks)

END

$R = \sqrt{6}$  ✓



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1. a) Divergence theorem states that "The total outward flux of vector field  $A$  enclosed in surface  $S$  is the same as the volume integral of the divergence of  $A$ ."

$$\oint A \cdot d\mathbf{s} = \int (\nabla \cdot A) dV$$

(b) Given,  $A = y \cos \phi \mathbf{a}_\rho + z \sin \phi \mathbf{a}_z$ .

from,  $\nabla \times A = \mathbf{a}_\rho \frac{\partial}{\partial \rho} + \mathbf{a}_\phi \frac{1}{\rho} \frac{\partial}{\partial \phi} + \mathbf{a}_z \frac{\partial}{\partial z}$

$$\nabla \times A = \frac{\partial}{\partial \rho} (y \cos \phi) + \frac{\partial}{\partial z} (z \sin \phi)$$

$$\nabla \times A = \cos \phi + \sin \phi$$

$$\int \nabla \times A \cdot d\mathbf{s} = \int \cos \phi d\rho + \int \sin \phi dz$$

$$= \rho \cos \phi + z \sin \phi \Big|_0^2$$

$$= 2 \cos \phi - 0 \sin \phi$$

$$= -2 \cos \phi \Big|_0^{60^\circ}$$

$$= -2 (\cos 60^\circ - \cos 0^\circ)$$

$$= -2 (-0.5)$$

$$\oint \nabla \times A \cdot d\mathbf{s} = 1$$

2. (a) Electrical potential is the work done per unit charge.  
 $V = W/Q$

(b) Given,  $Q_1 = -4 \mu C$  at  $(2, -1, 3)$  and  $(1, 0, 1)$ .  
from,  $V = \frac{k Q_1}{R} = \frac{Q}{4\pi\epsilon_0 R}$

$$R = \sqrt{(1-2)^2 + (0+1)^2 + (1-3)^2} = \sqrt{6}$$

$$R = \sqrt{6}$$

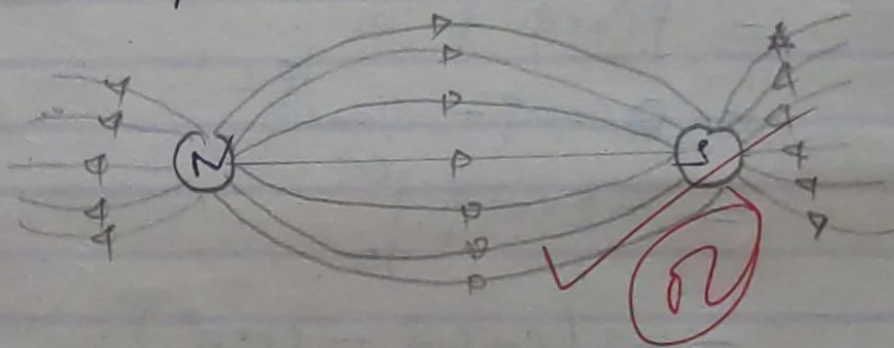
Then,



2. (b)  $V = \frac{-4 \mu C}{4\pi \epsilon_0 \times 6}$  but  $\epsilon_0 = 8.854 \times 10^{-12}$   
 $V = -1.468 \times 10^{10} \text{ J/C}$  (03)  
 $\therefore$  The potential is  $-1.468 \times 10^{10} \text{ J/C}$ .

3. (a), Coulomb's law state that "The electric flux  $\oint \vec{E} \cdot d\vec{A}$  through a closed surface  $S$  is equal to the total charge enclosed."  
 $\Phi = \oint \vec{E} \cdot d\vec{A} = \oint \rho_v dv$

3. (b) Electric field line pattern.



(c), Given,  $Q_1 = 1 \text{ mC}$  at  $(3, 2, -1)$ .

$Q_2 = 10 \text{ nC}$  at  $(0, 3, 1)$ .

~~$(0, 3, -1)$~~

$$(0, 3, 1) - (3, 2, -1) = (-3, 1, 2)$$

$$R^2 = (-3)^2 + 1^2 + 2^2$$

$$R^2 = 14$$

from,

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q_1 Q_2}{R^2}$$

$$F = \frac{1}{4\pi\epsilon_0} \cdot \frac{1 \text{ mC} \times 10 \text{ nC}}{14} = 6.42 \times 10^{-3} \text{ N}$$

$$F = 6.42 \times 10^{-3} \text{ N}$$

Then,

$$E = \frac{F}{Q}$$



$$(c) \quad E = \frac{6.42 \times 10^{-3} \text{ N}}{1 \text{ mC}}$$

$$E = 6.42 \text{ N/C}$$

"The force is  $6.42 \times 10^{-3} \text{ N}$  and electric field intensity is  $6.42 \text{ N/C}$ ."

(a) Coulomb's law states that "The force between two charges  $Q_1, Q_2$  is directly proportional to the product of their charge  $Q_1 Q_2$  and inversely to the square distance between them."

$$F \propto \frac{Q_1 Q_2}{R^2}$$

(a). Gauss's law states that "The electric flux  $\vec{v}$  of vector  $A$  through closed surface  $S$  is equal to the total charge enclosed"

$$Q = \oint \vec{v} \cdot d\vec{s} = \oint \rho \, dv$$

(b). (i). Charge enclosed  $= 15 \text{ nC} + 10 \text{ nC} - 20 \text{ nC}$   
 $= 5 \text{ nC}$

"The total charge enclosed is  $5 \text{ nC}$ ."

(ii) Total outward flux leaving the surface is  $5 \text{ wb}$ .